

Functional Requirements Document (FRD)

RetailMart Sales Performance Analytics Dashboard (Power BI)

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1. Project Overview

RetailMart requires an enterprise-grade, interactive Business Intelligence dashboard developed in Microsoft Power BI. The objective of this solution is to empower executive management, regional directors, and commercial analysts with deep, data-driven visibility into sales performance, profitability margins, customer segment behaviors, regional footprints, and long-term business growth trends. The dashboard will unify disparate retail transaction records into a single source of truth to support strategic planning and tactical operational adjustments.

2. Functional Requirements

Req ID	Functional Module	Description & System Specification
FR-01	Data Integration & ETL	The system must ingest transactional data from enterprise SQL databases, standardizing currency, date formats, and regional hierarchies automatically on a scheduled nightly refresh.
FR-02	Interactive Filtering	All dashboard pages must support real-time global and page-level filtering across geographical, temporal, and product dimensions with sub-2-second query render times.
FR-03	Dynamic Drill-Through	Management users must be able to right-click summary visual elements (e.g., Regional Sales) to drill through into granular transaction-level details.
FR-04	Role-Based Access Control	Power BI Row-Level Security (RLS) must restrict regional managers to view data strictly corresponding to their assigned geographic territories.

3. Dashboard Pages

Page Name	Primary Focus & Target Persona
Page 1: Executive Summary	High-level macro performance overview designed for C-level executives. Features core top-line KPIs, high-level growth trajectories, and regional revenue distribution summaries.
Page 2: Regional & Geographic Analysis	Geospatial performance tracking enabling regional directors to evaluate state-level revenue concentrations, profit leakages, and territory-specific order volumes.
Page 3: Product & Category Performance	Granular category and sub-category profitability analysis designed for merchandise managers to optimize inventory stocking and markdown strategies.
	Behavioral analytics tracking customer segment purchasing habits, order frequencies, Average Order Value (AOV), and lifetime value metrics.

Page Name	Primary Focus & Target Persona
Page 4: Customer Behavior & Segmentation	

4. Key Performance Indicators (KPIs)

KPI Name	DAX Formula / Calculation	Business Objective
Total Sales	SUM(Transactions[SalesAmount])	Measures total top-line revenue generated across all filtered business segments.
Total Profit	SUM(Transactions[Profit])	Tracks net financial earnings after deducting cost of goods sold (COGS) and discounts.
Profit Margin	DIVIDE([Total Profit], [Total Sales], 0)	Evaluates operational efficiency and pricing health as a percentage return on revenue.
Total Orders	DISTINCTCOUNT(Transactions[OrderID])	Quantifies transaction volume and operational throughput across retail channels.
Average Order Value (AOV)	DIVIDE([Total Sales], [Total Orders], 0)	Measures customer basket size monetization efficiency per transaction.
Customer Count	DISTINCTCOUNT(Transactions[CustomerID])	Tracks unique active buyer acquisition and consumer base expansion over time.

5. Dashboard Visuals

Visual Name	Business Purpose	Chart Type	Data Used (Dimensions & Measures)
Sales & Profit Growth Trend	Evaluates seasonal business growth trajectories and year-over-year revenue velocity.	Line & Stacked Column Chart	X: Year-Month Y (Bar): Total Sales Y (Line): Total Profit
Regional Revenue Distribution	Identifies top-performing geographic territories and expansion opportunities.	Filled Map / Shape Map	Location: State / Region Metric: Total Sales (Color saturation)
Category Profitability Matrix	Highlights high-margin vs. low-margin product lines to steer merchandising decisions.	Treemap	Group: Category Values: Total Sales Color Saturation: Profit Margin
Top 10 Sub-Categories	Pinpoints specific product drivers contributing most significantly to top-line revenue.	Horizontal Bar Chart	Y: Sub-Category X: Total Sales (Sorted Descending)
Customer Segment Contribution	Analyzes revenue distribution across	Donut Chart	Legend: Customer Segment Values: Total Sales

Visual Name	Business Purpose	Chart Type	Data Used (Dimensions & Measures)
	Consumer, Corporate, and Home Office segments.		
Payment Gateway Breakdown	Monitors preferred customer checkout methods and gateway transaction volumes.	Clustered Column Chart	X: Payment Method Y: Total Orders & Total Sales

6. Filter Specifications

Filter Name	Data Field Source	Behavior & Configuration Details
Region	Location[Region]	Global multi-select slicer (North, South, East, West). Controls regional filtering across all dashboard pages.
State	Location[State]	Cascading slicer dependent on Region selection; enables state-level deep dive.
Category	Product[Category]	Multi-select dropdown slicer (Furniture, Office Supplies, Technology).
Sub Category	Product[SubCategory]	Hierarchical multi-select slicer filtered dynamically by selected Category.
Customer Segment	Customer[Segment]	Checkbox filter allowing isolation of Consumer, Corporate, or SME accounts.
Payment Method	Transactions[PaymentMethod]	Checkbox filter (Credit Card, PayPal, Bank Transfer, Cash on Delivery).
Year	Calendar[Year]	Relative date or dropdown slicer covering historical operational years (2023–2026).
Month	Calendar[MonthName]	Multi-select calendar month filter for seasonal trend analysis.

7. User Actions & Interactivity

User Action	System Response & Expected Behavioral Outcome
Cross-Filtering (Clicking a Bar/Slice)	Clicking a specific category bar (e.g., "Technology") instantly cross-filters all adjacent visuals, KPIs, and charts on the page to isolate Technology transactions.
Drill-Through Execution	Right-clicking a state bar on the regional map and selecting <i>"Drill Through -> Detailed Transaction View"</i> redirects the user to a dedicated transaction ledger page filtered strictly for that state.
Slicer Synchronization	Slicers designated as "All Pages" automatically sync filter states when navigating between Executive Summary and specialized functional tabs.
Tooltips on Hover	Hovering over trend line data points surfaces a rich-text tooltip displaying exact Sales, Profit, and Profit Margin variance percentages against prior periods.

8. Acceptance Criteria

Ref ID	Requirement Area	Validation Criteria
AC-01	Data Reconciliation	Total Sales and Total Profit figures must reconcile 100% with audited enterprise SQL general ledger balances within a 0.00% rounding tolerance.
AC-02	Render Latency	Initial dashboard report load time and subsequent filter application must complete in under 2.0 seconds under standard concurrent user loads.
AC-03	DAX Calculation Accuracy	All measure calculations (AOV, Profit Margin, Customer Count) must correctly handle zero-division errors and filter context propagation.
AC-04	Drill-Through Integrity	Drill-through actions must successfully pass filter context without data leakage or context loss across pages.
AC-05	Row-Level Security (RLS)	Regional managers logging into Power BI Service must only view data corresponding to their authorized geographical territories.
AC-06	Mobile Responsiveness	The Power BI mobile layout view must render all core KPIs and summary cards cleanly on iOS and Android devices without text clipping.
AC-07	Export to PDF / Excel	Exporting dashboard pages to PDF must preserve layout integrity, formatting, and high-resolution chart rendering without truncation.
AC-08	Slicer Synchronization	Global slicers must persist selections when navigating between Executive, Regional, and Customer dashboard pages.
AC-09	Tooltip Formatting	All visual tooltips must display currency formatted to two decimal places and percentages formatted with explicit sign indicators (+/-).
AC-10	Automated Refresh	Scheduled nightly data gateway refreshes must execute successfully without throwing timeout or schema mismatch errors.

9. Assumptions

- Enterprise SQL data warehouse schema remains stable with primary and foreign key relationships intact between transaction and dimension tables.
- Management stakeholders are provisioned with Power BI Pro or Premium Per User (PPU) licenses required for workspace collaboration and scheduled refreshing.
- Historical transactional data prior to January 2023 is archived and excluded from active executive dashboard reporting models.

10. Constraints

- Power BI semantic model row counts must adhere to underlying import mode memory constraints (optimized for under 10M rows without enterprise Azure Analysis Services).
- Custom visual objects are strictly restricted to certified Microsoft AppSource visuals to ensure enterprise security compliance.
- Network bandwidth limitations in remote retail stores require optimized report payload sizes under 50MB.

11. Future Enhancements

- **Predictive Sales Forecasting:** Integrate Power BI AI Insights and machine learning regression models to forecast next-quarter revenue and inventory demand.
- **Automated Alerting:** Implement Power BI metric scorecards to trigger automated email/Teams alerts when profit margins drop below defined threshold limits.
- **Customer Churn Analytics:** Incorporate advanced clustering to identify dormant customer accounts and predict churn risk indicators.